

Depth-Damage Curve Model (MKM-DD-001)

Wolfram Oracle Benchmark of the Code and its v2.0 Document

MKM Research Labs

Version 0.2 — 03-November-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs ("MKM").

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an "as is" basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

Contents

1 Executive Summary **3**

2 Model Purpose and Scope **3**

2.1 What is benchmarked 3

2.2 What is not 4

3 Mathematical Framework **4**

4 Benchmark Method **4**

5 Validation and Backtesting **5**

5.1 The model document's claims 6

6 Model Limitations and Known Weaknesses **6**

7 Document History **6**

1 Executive Summary

This document benchmarks the Depth-Damage Curve Model (MKM-DD-001): the functions in `src/models/floodrisk/` that turn a flood at a gauge into depth, timing and damage at a property. It asks whether the code computes what its model document (`flood_risk_v2.0.pdf`) says, and whether what the document says holds.

Version 0.1 tested the filed v1.1 of that document and found it no longer described the model. The code was right, but the document gave a 2,000 m retention length where the code uses 10,000 m, described equations the code had dropped, and printed velocities, a curve-shift sensitivity and a distance error that its own equations do not give. The document has been rewritten to the code as v2.0, and the retention docstring corrected to the 10,000 m it uses. This version tests v2.0.

The code computes its documented formulae. The curve, Manning's velocity, slope, travel time, retention, haversine, IDW and correlation matrix agree with a 50-digit implementation of the document's equations to 3.2×10^{-16}. The property hydrograph agrees to the millimetre the code reports. The portfolio Monte Carlo's mean loss is within 1.82 standard errors of the exact expected loss (Table 1). Wolfram, computing each independently, agrees with the reference to 4.1×10^{-31}.

The document's claims hold: 15 of 15 (Table 2).

- All 70 figures in its sensitivity tables reproduce from the code, or, for the curve-shift table, from the exact expected loss.
- Its statements about what the code does are true: the retention length, retention applied once, the unread constants, the SVD draws and the 61% degree error.
- The fit it quotes for the degree-5 curve is exact.

Evidence. 14 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-03 by `python -m tools.wolfram_oracle dd001`.

2 Model Purpose and Scope

2.1 What is benchmarked

The functions of `src/models/floodrisk/` that v2.0 documents, imported as production imports them:

- `depth_damage.scalar_depth_damage`, the curve of record;
- `velocity`: Manning, slope, travel time, retention and the property hydrograph, which MKM-PF-001 calls for every property and storm;
- `spatial`: the haversine, IDW and the correlation matrix;
- `risk_analytics`: the Monte Carlo, VaR, ES and HHI.

2.2 What is not

- The calibration of the curve to survey data and the TUFLOW comparison (v2.0's §7.1–7.2), which are empirical.
- The propagation of MKM-PF-001 and the compound hydrograph of MKM-HG-001.
- KMeans clustering.

This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements v2.0's equations directly from its text:

- Table 1's curve, interpolated linearly (Eq. 1);
- Manning's $v = h^{2/3}S^{1/2}/n$ with $S \geq 0.001$, the slope and $t = d/(3600v)$ (Eqs. 3–5);
- the retention $e^{-d/\ell}$ (Eq. 6);
- the hydrograph of Eq. 7, which scales the gauge's shape to a peak that already carries the retention;
- the haversine (Eq. 8), IDW, and the correlation $\rho_0 e^{-111 d_{\text{deg}}/(L/1000)}$ (Eq. 9).

The expected loss, exactly. Eq. 10 shocks each damage ratio to $\text{clip}(\delta(1 + 0.2z), 0, 1)$ with z standard normal. Since $X = \delta(1 + 0.2z) \sim N(\delta, (0.2\delta)^2)$,

$$\mathbb{E}\,\text{clip}(X, 0, 1) = \delta\big[\Phi(\beta) - \Phi(-5)\big] + 0.2\delta\big[\varphi(-5) - \varphi(\beta)\big] + 1 - \Phi(\beta), \quad \beta = \tfrac{1-\delta}{0.2\delta}, \tag{1}$$

which is v2.0's Eq. 11. The expected portfolio loss is $\sum_j V_j\, \mathbb{E}\,\text{clip}(X_j, 0, 1)$ whatever the correlation, and it gives the curve-shift table exactly.

4 Benchmark Method

The cases.

- Every family is evaluated over a grid covering the document's tables and their edges: depths from -0.2 to 7 m, zero and negative roughness, slopes below the floor, and negative distances.
- The hydrograph is run on a 168-hour triangular storm at three travel times.
- The Monte Carlo runs on a 40-property portfolio near the Thames, with depths spanning every damage band. It uses the production defaults: 1,000 draws, seed 42, and the code's own correlation matrix.

The printed tables. Tables 4–6 and 8 are judged as printed: the code's value, rounded and formatted as the house generator does, must give each printed figure.

The statements. Each statement about the code is checked against the code:

- that retention is applied once, by running the hydrograph with different retention arguments;
- that the constants are unread, by searching `src`;
- that the draws are by SVD, by reproducing NumPy's legacy draws from the SVD factor.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the curve, through its own first-order `Interpolation` of Table 1;
- Manning and the retention;
- the great-circle distance, as R times the angle between the two points' unit vectors;
- the clipped expectation, by `Expectation`;
- the least-squares fit of a degree-5 curve through the origin to Table 1;
- the true length of 0.1° east–west at 51.5°N, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eq. 1, the depth-damage curve: code against the reference, worst relative error	2.2×10^{-16}	1.0×10^{-12}	pass
Eq. 3, Manning's velocity: code against the reference, worst relative error	1.9×10^{-16}	1.0×10^{-12}	pass
Eq. 4, the ground slope: code against the reference, worst relative error	5.6×10^{-17}	1.0×10^{-12}	pass
Eq. 5, the travel time: code against the reference, worst relative error	2.9×10^{-16}	1.0×10^{-12}	pass
Eq. 6, the retention: code against the reference, worst relative error	3.2×10^{-16}	1.0×10^{-12}	pass
§3.4.2, IDW: code against the reference, worst relative error	8.2×10^{-17}	1.0×10^{-12}	pass
Eq. 5: no velocity gives an infinite travel time	yes		pass
Eq. 8, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eq. 9, the correlation matrix: worst relative error	3.1×10^{-16}	1.0×10^{-12}	pass
Eq. 7, the property hydrograph: worst error (m)	7.5×10^{-5}	1.0×10^{-4}	pass
Eqs. 10–11: Monte Carlo mean loss (1,000 draws) against the exact, $\lvert z \rvert$	1.82	4	pass
§3.5: the code's ES_{99} against its own draws	0	1.0×10^{-12}	pass
Wolfram agrees with the reference	4.1×10^{-31}	1.0×10^{-25}	pass
§3.1: the degree-5 curve is the exact least-squares fit to Table 1	4.4×10^{-9}	5.0×10^{-9}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
Table 4	the depth-damage curve at 13 depths	13 of 13 figures reproduced	met
Table 5	Manning's velocity	18 of 18 figures reproduced	met
Table 6	retention by distance and length	24 of 24 figures reproduced	met
Table 7	expected loss after a ±10% curve shift, by damage ratio	7 of 7 figures reproduced from Eq. 11	met
Table 8	velocity and travel time by roughness	8 of 8 figures reproduced	met
§3.3.4, Table 2	retention length $\ell = 10{,}000$ m	the code's default is 10,000 m	met
§3.3.4, §6.3	94, 82, 61% retained at 0.6, 2, 5 km; 0.37, 0.67, 0.82 at 2 km	as printed	met
§3.3.5	retention is applied once, by the caller, not in the hydrograph	the code's hydrograph is Eq. 7 without it, to 7.5×10^{-5} m	met
§3.1, §3.2	the property-type factors and 25 km radius are unread	`PROPERTY_TYPE_DAMAGE_FACTORS`, `FLOOD_MAX_DISTANCE_M` never read in `src`	met
§3.4.3, §8(7)	1° ≈111 km overstates east–west distance by 61% at 51.5°N	60.6%; on WGS84, 0.1° east–west is 6,944 m, not 11,100	met
§3.4.3	the correlation matrix is positive definite	smallest eigenvalue 0.567	met
§3.5	shocks drawn by SVD	NumPy's legacy draws equal the SVD factor's exactly	met
§3.5	HHI $= 1/N$ for an equal portfolio	0.0250 for $N = 40$	met
Q3	800 MB for 10,000 properties	800 MB	met
§3.1	degree-5 curve: R^2 0.999244, RMSE 0.009368, largest error 0.020388, increasing	0.999244, 0.009368, 0.020388; least slope 0.052	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented formulae. Whether the curve fits UK damage, and whether 10,000 m is the right retention length, are empirical questions; v2.0 makes the latter a condition of its Q9.

2. **Unused code.** IDW, the correlation matrix, the portfolio analytics and the degree-5 curve are verified, but nothing in production calls them; the property-type factors and 25 km radius sit unread in config.

7 Document History

Date	Version	Description	Author
03-Nov-2026	0.1	First issue: tests the filed v1.1; finds it no longer describes the model (not filed)	David K Kelly
03-Nov-2026	0.2	Tests v2.0, rewritten to the code	David K Kelly